Week - 1
Skill builder

1) Input 1: 90
80

output: The integer closer to 100 is 90

Code:
```
import java.util.*;
class main{
public static void main(String[] args)
{
 Scanner sc = new Scanner(system.in);
 int a = sc.nextInt();
 int b = sc. nextInt();
 x = Math.abs(100-a);
 y = math.abs (100-b);
 if(x<y){
 { System.out.println("a is closer");
 else {
  System.out.println("b is closer");
 }}
}
```

2) Input 1:
4
3

output:
one is ok but other is not divisible by 3.

code:
```
import java.util.*;
class main{
public static void main(String[] args){
{ Scanner sc = new Scanner (System.in);
 int a = sc.nextInt();
 int b = sc.nextInt();
 if(((a>0) && (b%3!=0))||((b>0) && (a%3!=0))){
 System.out.println("one is ok but other is not % by 3");}
 else {
  System.out.println("neither is ok");
 }
 }
}
```

20 Original integer : 20
Converted double : 20.0

Code:
```
import java.util.*;
class main {
public static void main (String[] args){
 Scanner sc =new Scanner(System.in);
 int a= sc.nextInt();
 double n= a;
 System.out.println("Original :"+ a);
 System.out.println("Converted : " +n);
 }
}
```

4) Input:
1 2

output.
Sum is not multiple of product

Code:
```
import java.util.*;
class main {
public static void main (String[] args)
Scanner sc = new Scanner (system.in);
int a= sc.nextInt();
int b= sc.nextInt();
int sum,pro;
 sum= a+b;
 pro= a*b;
 if (sum==pro) {
  System.out.println("sum is equal to product");}
 else {
  System.out.println("sum is not equal to product");}
 }
}
```

5) Input:
3.0

output:
circumference : 18.85
area : 28.27

Code:
```
import java.util.*;
class main {
public static void main (String[] args){
Scanner sc =new Scanner(System.in);
double r= sc.nextFloat();
double c,a;
 c= 2*3.14159*r;
 a= 3.14159*r*r;
 System.out.printf("circum: %.2f meters\n");
 System.out.printf("area: %.2f squaremeters\n");
}}
```

6) Input:
85
2

output:
Result : 1

Code:
```
import java.util.scanner;
class main {
public static void main (String[] args){
Scanner sc= new Scanner (system.in);
int n= sc.nextInt();
int m= sc.nextInt();
int res= n & ((1<<m)-1);
system.out.println("Result: "+res);
}
}
```

Week 2 SB

1) Input:

50 60 70 80 70

output:

avg : 70

~~passed~~ passed.

code:

```
import java.util.*;
class main {
public static void main (String[] args){
Scanner sc =new Scanner (System.in);
int a= sc.nextInt();
int b= sc.nextInt();
int c= sc.nextInt();
int d= sc.nextInt();
int e= sc.nextInt();
int avg= (a+b+c+d+e)/5 ;
if (avg >=50 ){
System.out.println ("passed");
else {
System.out.println("fail");
}
}
```

2) Input:

10

output:

10 is a multiple of 5

code:

```
import java.util.Scanner;
class main{
public static void main (String[] args){
Scanner sc= new Scanner(System.in);
int a= sc.nextInt();
if((a%5)==0)
    System.out.printf("%d is a multiple of 5",a);
}
}
```

3) Input:

1.2

45.2

output:

BMI : 31.37

type : obese

code:

```
import java.util.Scanner;
class main{
public static void main (String[] args){
Scanner sc =new Scanner ( System.in );
double h= sc.nextDouble();
double w= sc.nextDouble ();
double bmi= w/(h*h);
System.out.println("type");
if(bmi <18.5)
    System.out.println("underweight");
else
    System.out.println ("obese");
}
}
```

4) Input

20000

5

output

current value : 8874.11

category : medium

code:

```
import java.util.Scanner;
class main{
public static void main ( String[] args){
int a = sc.nextInt();
int b = sc.nextInt();
double c= a*Math.pow(0.85,b);
System.out.printf("current value: %.2f \n", c);
if (c>100000)
    System.out.println("category : High");
else
    System.out.println("category : low");
}
}
```

5) Input:
20

output:
the no: of digits in 20 matches the sum of its digits.

code:

```
import java.util.Scanner;
class main{
public static void main (String[] args){
 Scanner sc = new Scanner( System.in);
 int a = sc.nextInt();
 int sum=0, temp=a, count=0;
 do{
   int digit = temp % 10;
   sum+=digit;
   temp/=10;
    count++;
  } while(temp!=0)
 if (sum == count)
    System.out.println ("same");
 else
    System.out.println("not same");
```

6) Input:
3

output:

```
*
* *
* * *
* *
*
```

code:

```
import java.util.scanner;
class main {
public static void main (String[] args){
int a= sc.nextInt();

//logic
}
}
```

Week - 3 SR

1) Input:
3
10 28 47

output.
38

code:

```
import java.util.*;
class main{
 public static void main (String [] args){
 Scanner sc = new Scanner (System.in);
 int n= sc.nextInt();
 if (n<3){
        return;}
 int a[]= new int[n];
 for (int i=0; i<n; i++){
    int num = sc.nextInt();
 }  a[i] = num;
 Arrays.sort(a);
 int  sum = a[1] + a[n-3];
 System.out.println(sum);
}}
```

2) Input:
3
1 2 3
4 5 6
7 8 9

output:
sum of main diagonal: 15
sum of secondary diagonal: 15

Code:

```
import java.util.scanner;
Class main{
public static void main (String[] args){
 Scanner sc = new Scanner( System.in);
 int n= sc.nextInt();
 int a[][]= new int [n][n];
 int m_sum=0, ssum=0;
```

```
for(int i=0; i<n ; i++){
    for( int j=0 ; j<n; j++){
        int num =sc.nextInt();
        a[i][j] =num;
        if (i==j){
            msum= a[i][j] ;
        }
    }
}

for (int i=0 ; i<n ; i++ ){
    ssum = a[i] [n-1-i] ;}

System.out.println ( msum, ssum); 
}
}
```

3) Input:

5
10 20 30 40 50

output:

60

```
import java.util . scanner ;
class main{
public static void main (String[] args){
  Scanner sc =new Scanner (System.in );
  int n=sc.nextInt() ;
    a[i] =num;
  }
  int sum= a[0] + a[n-1];
  System.out.println (sum);
}
}
```

Week - 4 SB

1) Input:

1
Hello, world.

output:

2 1

Code:

```
import java.util.scanner ;
class main {
  public static void main (String[] args ){
  Scanner sc= new Scanner (System.in );
  int a = sc .nextInt();
  sc.nextLine() ;
  for (int j=0, j<s.length(); j++){
    char c = s. charAt (j);
    if( c== ',') {
          c1++ ; }
    else if ( c== '.'){
      c2++; }
  }
}
```

```
2) import java.util.* ;
class main{
  public static void main (String args[]){
  Scanner sc=new Scanner (system.in);
  int n= sc.nextInt();
  sc.nextLine();
  String sentence = sc.nextLine();
  String[] words = sentence . split("\\s");
  Arrays . sort (words);
  for (int i=0; i<n ; i++){
  System. out. print( words [i]);
  if (i<n-1){
    system.out. print(" ");
  }
}
}
}
}
```

3)
```
import java.util.*;
class main{
public static void main (String args[]){
Scanner sc=new Scanner (System.in);
sc.nextLine();
for (int i=0 ;i<T; i++){
  String s= sc.nextLine();
  if s.length()==10 && s.charAt(0)!='0' && s.matches
  ("[0-9]+"){ System.out.println ("Yes");
  else{ System.out.println("No");
}}}
}
```

4)
```
import java.util.*;
class main{
public static void main (String args[]){
Scanner sc=new Scanner(System.in);
String sentence =sc.nextLine();
String[] words = sentence.split("\\s+");
boolean found = false;
for (String word : words){
 if( word.matches("[a-zA-Z]+") && word.length()>=2)
{
  System.out.println (word+" ");
  found =true;
 }
}
if (!found){
  System.out.println("No valid words");
}}
```

Week-5

1)
```
import java.util.*;
class Bank{
public static void main (String args[]){
Scanner sc=new Scanner(System.in);
int n= sc.nextInt();
sc.nextLine();
for(int i=0; i<n; i++){
  int accno= sc.nextInt();
  sc.nextLine();
  String name = sc.nextLine();
  double bal = sc.nextDouble();
  double deposit= sc.nextDouble();
  double withdraw= sc.nextDouble();
  Bank b =new Bank(accno,name,bal);
  b.deposit(deposit);
  b.withdraw(withdraw);
  System.out.println("Account Number: "+b.getaccno());
  System.out.println("customer Name: "+b.getname());
  System.out.printf("Final Balance: %.1f",b.getbal());
 }
}
```

2)
```
import java.util.*;
class Electricity {
 private int id;
 private String name;
 private double units;
 Electricity( int id, String name, double units){
 this.id = id;
 this.name = name;
 this.units = units; }
 public static void main (String args[]){
 Scanner sc=new Scanner (System.in);
 int n= sc.nextInt();
 Sc.nextLine();
 for (int i=0; i<n; i++){
  int id =sc.nextInt();
  String name =sc.nextLine();
  double units = sc.next Double();
}}
```

3)
```
import java.util.scanner;
class Booking {
    private int id;
    private string name;
    private double distance;
    private booking ( int id, string name, double distance){
    this.bookingid = id;
    this.name = name;
    this.distance = distance;
    calculate Fare();
}
class cityapp {
public static void main( String[] args){
 scanner sc = new scanner (system.in);
 int n = Integer. parseInt (sc.nextLine());
 string name = sc.next Line();
 double distance = Double.parse Double (sc.nextLine());
 Booking booking = new Booking (id, name, distance);
 System.out.printf("Final Fare : %.1f\n", booking.
                                  getFare());
 }
}
}
```

Week-6

1)
```
class premium subscription{
    double base;
    double service;
    double cost;
    premium subscription (double b, double s, double c){
     base = b;
     service = s;
     cost = c; }
   double calculate Monthly Cost() {
       return base + service + cost;
   }
}
```

2)
```
class product {
    public double price;
    product (double price){
      this.price = price;
    }
}
class DiscountedProduct extends Product {
  Discounted Product (double price, double rate) {
   super (price);
   this.rate = rate;
  }
  private double rate
  double calculate Selling price() {
     return price * (1-rate);
  }
}
```

3)
```
class sales Tax Calculator {
    static int calculate Final Price (int price, int tax){
    return price + ((price * tax) /100);
    }
    static double calculateFinalPrice (double price,
                                     double tax ){
    return price + ((price * tax )/100);
}}
}
```

4)
```
class cuboid{
    double length, width, height;
    cuboid (double l, double w, double h){
        length = l;
        width = w;
        height = h;
    }
    double calculateVolume(){
        return length * height * width;
    }
}

class cube extends cuboid{
    cube(double s){
        super(s,s,s);
    }
    public double calculateVolume(){
        return Math.pow(length, 3);
    }
}
```

5)
```
class Item{
    protected String name;
    protected double price;
    public Item (String name, double price){
        this.name = name;
        this.price = price;
    }
    public double calculateCost(){
        return price;
    }
}
class produce extends Item{
    public produce (String name, double price){
        super (name, price);
    }
    @override
    public double calculatecost(){
        return price;
    }
}
```

Week-7

1)
```
interface costcalculator{
    void getEnergyDetails (Scanner scanner)
    void calculateAndDisplayCost();
}
class EnergyConsumptionTracker implements
    costcalculator{
        double ratePerUnit;
    int numDays;
    double[] dailyConsumption;
    EnergyConsumptionTracker(double ratePerUnit
            int numDays){
        this.ratePerUnit = ratePerUnit;
        this.numDays = numDays;
        this.dailyConsumption = new Double[numDays];
    }
    public void getEnergyDetails (Scanner scanner){
    for(int i=0; i<n; i++){
        dailyConsumption = new Double[numDays];
    }
    public void getEnergyDetails (Scanner scanner){
        System.out.printf("Day %d: Rs %.2f/n", i+1,
                                        daycost);
    }
    System.out.printf("total energy cost: Rs%.2f/n",
                                totalcost);
}
```

2)
```
interface healthcalculator{
    double calculateBMI(double weight, double
                                height);
}
class BMICalculator implements healthcalculator{
public double calculateBMI (double weight, double
                                height);
{
    if (weight <=0 || height <=0){
                return -1; }
    } return weight/(height*height);
}
```

3) interface interestcalculator{

```
double simpleinterest (double principal, double
                       rate, int time);
}
class simpleinterestcalculator implements interest
                                  calculator{
  public double simpleinterest (double principal,
                    double rate, int time){
     return(principal * rate * time)/100;
  }
}
```

4) Input: output :

1934 you are 90 yrs old

Code :

```
interface agecalculator{
    int calculateage(int birthyear);
}
class ...manage calculator implements agecalculator{
    private static final int current-year = 2024;
  public int calculateage ( int birthyear){
      return current-year - birth year;
  }
}
```

5) interface inventory{

```
    void add product ( string name, double price, int
                                quantity);
    double calculatetotalvalue();
}
class product{
  private string name;
  private double price;
  private int quantity;
public product (string name, double price, int quantity){
     this.name = name;
     this price = price;
     this.quantity = quantity; }
  public double getvalue(){
      return price * quantity
  }
}
class simple inventory implements inventory{
  private product[] products;
  private int count;
  public simple inventory (int capacity){
     products = new product [capacity];
     count =0;
  }
}
public void add product ( string name, double price,
                          int quantity){
  if (count < products. length){
     products [count ++] = new product( name, price,
                                   quantity);
        system.out. println ("product added to
                            inventory.");
     }
  }
  public double calculatetotalvalue(){
     double total = 0.0;
     for(int i=0; i< count; i++){
        total += products [i]. getvalue();
     }
     return total;
  }
}
```

Week. 8

1)
```
import java.util.*;
class Invalid Duration Exception extends Exception{
    Invalid Duration Exception (String msg){
        super (msg);
    }
}
public class main{
    public static void main (string args[]){
        scanner sc= new scanner (system.in);
        int n= sc.next Int();
        try{
            validate Meeting Duration (n);
            system.out.println("Meeting scheduled
                                 successfully");
        }
        catch{ Invalid Duration ( Exception e )}{
            system.out.println("Error");
        }
    }
    static void validate Meeting Duration (int n) throws
    Invalid Duration Exception{
        if ( n<=0|| n>240) {
            throw new Invalid Duration Exception
                ("Invalid meeting duration");
} } }
```

2)
```
public class main{
    public static void main (string args[]){
        scanner sc= new scanner (system.in);
        string username = sc.nextLine();
    try{
        validate username (username );
        System.out.println("username is valid: ");
    }
    catch ( Invalid username Exception e)
        system.out.println("Invalid username");
    } }
    static void validateusername (string username ) throws
     Invalid usernome exception{
        if (username.contains(" "))
            throw new invalid usernome exception
                 ("username cannot contain space");

        if (username. length() <5)
            throw new invalid usernameexception(" username
                 must be atleast 5 char);
}
```

3)
```
public class main{
    public static void main (string args[]){
        scanner sc = new scanner( system.in);
        string filename = sc.nextLine();
        try{
            validate Filename (filename);
            System.out.println("valid file name ");
        }
        catch (Invalid File Name Exception e){
            system.out.println("Error");
        }
    }
    static void validate Filename (String filename)
    throws Invalid filename exception{
        if ( filename .length() <3 || !filename.matches
                ("[A-Za-z0-9] *")){
            throw new invalidnameexception("invalid");
        }
    }
}
```

Week-7.

1)
```
import java.util.*;
class main{
    public static void main (string args[]){
        scanner sc = new scanner(system.in);
        int n = sc.nextInt();
        for (int i=0; i<n; i++){
            int num = sc.nextInt();
            if (list.isEmpty() || num > list.get(list.size() -1)){
                list.add(num);
            }
        }
        system.out.print(list);
    }
}
```

2)
```
import java.util.*;
class main{
public static void main (string args[]){
    scanner sc = new scanner(system.in);
    int n = Integer.parseInt(sc.nextLine());
    LinkedList<string> playlist = new LinkedList<>();
    int currentIndex = 0;
    for( int i=0; i<n; i++)
        string command = sc.nextLine();
    if (command.startswith("ADD")){
        string song = command.split(" ")[1];
        playlist.add(song);
        if (playlist.size() == 1){
            currentIndex = 0;
        }
    }
else if (command.startswith("Remove")){
    string song = command.split(" ")[1];
    int removeIndex = playlist.indexOf(song);
    if (removeIndex != -1){
        playlist.remove(removeIndex);
        if (playlist.isEmpty()){
            currentIndex = 0;
        }
        else if (removeIndex < currentIndex){
            currentIndex--;
        }
        else if (currentIndex >= playlist.size()){
            currentIndex = 0;
        }
    }
}
else if (command.equals("SHOW")){
    if (playlist.isEmpty()){
        system.out.println("EMPTY");
    }
    else{
        for (string s : playlist){
            system.out.print(s+" ");
        }
        system.out.println();
    }
}
else if (command.equals("NEXT")){
    if (playlist.isEmpty()){
        system.out.println("EMPTY");
    }
    else{
        currentIndex = (currentIndex +1)
            % playlist.size();
        system.out.println(playlist.get(currentIndex));
}}}}}
```

Week-10

1) import java.util.*;

```
class vehicle{
    string regNumber;
    string ownerName;
    string vehicleType;
    public vehicle(string regNumber, string ownerName,
        this.regNumber = regNumber;
        this.ownerName = ownerName;
        this.vehicleType = vehicleType;
    }
    public boolean equals(object obj){
    if (this == obj) return true;
    if ((obj == null) || getclass() != obj.getclass())
        return false;
        vehicle v = (vehicle) obj;
        return regNumber.equals(v.regNumber);
    }
public int hashCode(){
    return regNumber.hashCode();
  }
  public string tostring(){
    return regNumber +" "+ ownerName +" "+
            vehicleType;
    }
}
public class Main{
    public static void main(string args[]){
    scanner sc = new scanner(system.in);
    int n = sc.nextInt();
    sc.nextLine();
    HashSet<vehicle> vehicleset = new HashSet<>();
  for(int i=0; i<n; i++){
        string regNumber = sc.next();
        string ownerName = sc.next();
        string vehicletype = sc.next();
        vehicle v = new vehicle(regNumber, ownerName,
                        vehicleType);
        vehicleset.add(v);
    }
}}}
```

2) import java.util.*;

```
class main{
    public static void main(string[] args)
    {
        scanner scanner = new scanner(system.in);
        int n = scan.nextInt();
        Treeset<Integer> s = new Treeset<>();
        for(int i=0; i<n; i++)
        {
            int a = scan.nextInt();
            s.add(a);
        }
        int find = sc.nextInt();
        if (s.contains(find)){
            system.out.println(find +" "+ is present!");
        }
        else{
            system.out.println(find +" is not present!");
} } }
```

input:

4
2 4 5 6
5

output:

5 is present!